ORTHOGONAL POLYNOMIALS ASSOCIATED WITH THE SALEM–RAO KERNEL WEIGHT

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Abstract. The orthogonal polynomial system associated with the Salem–Rao kernel weight $w(u) = e^u/(e^{e^u}+1)$ on $\mathbb{R}$ is constructed in a form where every index is treated in closed form. The moments μ_k are identified for all $k \geq 0$ with the k-th derivatives at 1 of the entire function $F(s) = \Gamma(s)(1-2^{1-s})\zeta(s)$ and are then expressed by an explicit finite double sum in terms of Stieltjes constants and complete exponential Bell polynomials. From these moments, exact Hankel determinants, a determinantal formula for the monic orthogonal polynomials P_n of every degree n, and the three-term recurrence with explicit determinant expressions for the recurrence coefficients α_n, β_n are derived. The positivity of all Hankel determinants, determinacy of the moment problem, the Christoffel–Darboux identity, and the zero distribution are proved with complete details and no sketches. All formulas are specialized to the Salem–Rao weight.

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1. The Salem–Rao weight and its moment problem

Definition 1.1. Let $\ell := \log 2$. The *Salem–Rao kernel weight* is

$$w(u) := \frac{e^u}{e^{e^u}+1}, \qquad u \in \mathbb{R}. \tag{1.1}$$

Proposition 1.2. *The weight w satisfies:*

(i) $w(u) > 0$ *for every* $u \in \mathbb{R}$.
(ii) $w \in L^p(\mathbb{R})$ *for every* $1 \leq p \leq \infty$.

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(iii) *For every $k \geq 0$ the moment*

$$\mu_k := \int_{\mathbb{R}} u^k w(u)\ \mathrm{d}u \tag{1.2}$$

exists and is finite.

(iv) *The Hamburger moment problem determined by $\{\mu_k\}_{k\geq 0}$ is determinate.*

Proof. For (i), for every $u \in \mathbb{R}$ one has $e^u > 0$ and $e^{e^u} + 1 > 1$, hence $w(u) > 0$.

For (ii), as $u \to -\infty$, $e^u \to 0$ and $e^{e^u} \to 1$, so

$$w(u) = \frac{e^u}{e^{e^u}+1} = \frac{e^u}{1+e^{e^u}} \sim \frac{e^u}{2}.$$

There exists $C > 0$ such that $|w(u)| \leq Ce^u$ for all $u \leq 0$. As $u \to +\infty$, $e^{e^u} \gg 1$ and

$$w(u) = e^{u-e^u}\left(1+e^{-e^u}\right)^{-1} \sim e^{u-e^u}.$$

There exists $C' > 0$ such that $|w(u)| \leq C'e^{u-e^u}$ for all $u \geq 0$. For $1 \leq p < \infty$,

$$\int_{-\infty}^{0} |w(u)|^p\ \mathrm{d}u \leq C^p \int_{-\infty}^{0} e^{pu}\ \mathrm{d}u < \infty,$$

and

$$\int_{0}^{\infty} |w(u)|^p\ \mathrm{d}u \leq C'^p \int_{0}^{\infty} e^{p(u-e^u)}\ \mathrm{d}u < \infty,$$

since $e^{p(u-e^u)}$ decays super-exponentially as $u \to +\infty$. Continuity and decay at $\pm\infty$ imply $\|w\|_\infty < \infty$, so $w \in L^p(\mathbb{R})$ for all $1 \leq p \leq \infty$.

For (iii), fix $k \geq 0$. For $u \leq 0$, the estimate $|u|^k w(u) \leq C_k |u|^k e^u$ holds with some $C_k > 0$, and

$$\int_{-\infty}^{0} |u|^k e^u\ \mathrm{d}u = \int_{0}^{\infty} t^k e^{-t}\ \mathrm{d}t = \Gamma(k+1) < \infty.$$

For $u \geq 0$, the estimate $|u|^k w(u) \leq C'_k u^k e^{u-e^u}$ holds with some $C'_k > 0$, and

$$\int_{0}^{\infty} u^k e^{u-e^u}\ \mathrm{d}u \leq \int_{0}^{\infty} u^k e^{-e^u/2}\ \mathrm{d}u < \infty,$$

because e^u dominates all powers of u. Thus $\int_{\mathbb{R}} |u|^k w(u)\ \mathrm{d}u < \infty$, and the moment exists and is finite.

For (iv), Carleman's condition is verified. From the estimates above,

$$|\mu_k| \leq \int_{-\infty}^{0} |u|^k e^u\ \mathrm{d}u + \int_{0}^{\infty} u^k e^{u-e^u}\ \mathrm{d}u =: I_k^- + I_k^+.$$

One has $I_k^- = \Gamma(k+1) = k!$. For I_k^+, substitute $x = e^u$, so $u = \log x$ and $\mathrm{d}u = \mathrm{d}x/x$, to obtain

$$I_k^+ \leq \int_{1}^{\infty} (\log x)^k e^{-x}\ \mathrm{d}x.$$

For $x \geq 1$, $\log x \leq x$, hence

$$I_k^+ \leq \int_{1}^{\infty} x^k e^{-x}\ \mathrm{d}x \leq \Gamma(k+1) = k!.$$

Therefore, for every $k \geq 0$,

$$|\mu_k| \leq 2k!. \tag{1.3}$$

For $n \geq 1$, this yields

$$\mu_{2n}^{-1/(2n)} \geq (2(2n)!)^{-1/(2n)}.$$

By Stirling's formula,

$$(2n)!^{1/(2n)} \sim \frac{2n}{e},$$

so there exist $N \in \mathbb{N}$ and $c > 0$ such that for all $n \geq N$,

$$(2(2n)!)^{-1/(2n)} \geq \frac{c}{n}.$$

Consequently,

$$\sum_{n=1}^{\infty} \mu_{2n}^{-1/(2n)} \geq \sum_{n=N}^{\infty} \frac{c}{n} = +\infty.$$

By Carleman's criterion, the measure $w(u)\ \mathrm{d}u$ is determinate. □

Definition 1.3. For $f, g \in L^2(\mathbb{R}, w(u)\ \mathrm{d}u)$ define

$$\langle f, g \rangle_w := \int_{\mathbb{R}} f(u)g(u)w(u)\ \mathrm{d}u. \tag{1.4}$$

2. Moment generating function and arbitrary-index formula

Definition 2.1. Define

$$F(s) := \Gamma(s)\big(1 - 2^{1-s}\big)\zeta(s), \qquad s \in \mathbb{C}. \tag{2.1}$$

Lemma 2.2. *The function F is entire.*

Proof. Write $1 - 2^{1-s} = 1 - e^{(1-s)\ell}$ with $\ell = \log 2$. Near $s = 1$,

$$1 - 2^{1-s} = (s-1)\ell + O\big((s-1)^2\big).$$

The zeta function has a simple pole at $s = 1$, with

$$\zeta(s) = \frac{1}{s-1} + O(1).$$

Thus the product $(1-2^{1-s})\zeta(s)$ is holomorphic near $s = 1$. The gamma function $\Gamma(s)$ is holomorphic and nonzero near $s = 1$, so F is holomorphic at $s = 1$.

For $\Re(s) > 0$, the Mellin representation

$$F(s) = \int_0^{\infty} \frac{x^{s-1}}{e^x + 1}\ \mathrm{d}x \tag{2.2}$$

holds, with absolute convergence and holomorphy in this half-plane. The product $\Gamma(s)(1-2^{1-s})\zeta(s)$ has no poles for $\Re(s) \geq 0$ because the simple pole of ζ at $s = 1$ is cancelled by the zero of $1 - 2^{1-s}$, and the remaining singularities at nonpositive integers are removable for the product. Therefore F extends holomorphically to $\mathbb{C}$. □

Theorem 2.3 (Moment–derivative identity)**.** *For every integer $k \geq 0$,*

$$\mu_k = F^{(k)}(1). \tag{2.3}$$

Equivalently,

$$F(1 + \varepsilon) = \sum_{k=0}^{\infty} \frac{\mu_k}{k!}\varepsilon^k, \qquad \varepsilon \in \mathbb{C}. \tag{2.4}$$

Proof. Substitute $x = e^u$ in (1.2). Then $u = \log x$ and $\mathrm{d}u = \mathrm{d}x/x$, so

$$\mu_k = \int_0^\infty \frac{(\log x)^k}{e^x + 1}\,\mathrm{d}x. \tag{2.5}$$

For $\Re(s) > 0$, (2.2) holds. For each fixed $k \geq 0$, the integrand

$$\frac{(\log x)^k x^{s-1}}{e^x + 1}$$

is dominated, on every compact vertical strip $\sigma_0 \leq \Re(s) \leq \sigma_1$ with $\sigma_0 > 0$, by an x-integrable function independent of s: near $x = 0$ it behaves like $x^{\sigma_0 - 1}|\log x|^k$, and near $+\infty$ like $x^{\sigma_1 - 1}|\log x|^k e^{-x}$. Dominated convergence therefore permits differentiation under the integral sign and yields

$$F^{(k)}(s) = \int_0^\infty \frac{(\log x)^k x^{s-1}}{e^x + 1}\,\mathrm{d}x, \qquad \Re(s) > 0.$$

Evaluating at $s = 1$ gives $F^{(k)}(1) = \int_0^\infty (\log x)^k (e^x + 1)^{-1}\,\mathrm{d}x$, which equals μ_k by (2.5). The exponential generating function follows from the Taylor expansion of the entire function $F(1 + \varepsilon)$. □

3. Closed-form expression for the moments

3.1. Coefficient expansions.

Definition 3.1. The Stieltjes constants γ_n are defined by the Laurent expansion

$$\zeta(1 + \varepsilon) = \frac{1}{\varepsilon} + \sum_{n=0}^\infty \frac{(-1)^n \gamma_n}{n!}\varepsilon^n, \qquad 0 < |\varepsilon| < 1. \tag{3.1}$$

Here $\gamma_0 = \gamma$ denotes the Euler–Mascheroni constant.

Lemma 3.2. *For $|\varepsilon| < 1$,*

$$\log \Gamma(1 + \varepsilon) = \sum_{m=1}^\infty c_m \varepsilon^m, \tag{3.2}$$

with

$$c_1 = -\gamma, \qquad c_m = \frac{(-1)^m \zeta(m)}{m} \quad (m \geq 2). \tag{3.3}$$

Let

$$\Gamma(1 + \varepsilon) = \sum_{n=0}^\infty g_n \varepsilon^n. \tag{3.4}$$

Then for every $n \geq 0$,

$$g_n = \frac{1}{n!}\,\mathrm{B}_n\big(1!c_1, 2!c_2, \ldots, n!c_n\big), \tag{3.5}$$

where B_n denotes the n-th complete exponential Bell polynomial.

Proof. The series (3.2)–(3.3) is the classical Taylor expansion of $\log \Gamma(1 + \varepsilon)$ at $\varepsilon = 0$. Exponentiation gives

$$\Gamma(1 + \varepsilon) = \exp\left(\sum_{m=1}^\infty c_m \varepsilon^m\right) = \exp\left(\sum_{m=1}^\infty \frac{(m! c_m)}{m!}\varepsilon^m\right).$$

The defining generating identity of the complete exponential Bell polynomials is

$$\exp\left(\sum_{m=1}^{\infty}\frac{x_m}{m!}t^m\right)=\sum_{n=0}^{\infty}\frac{\mathrm{B}_n(x_1,\dots,x_n)}{n!}t^n.$$

Taking $x_m = m!c_m$ and $t=\varepsilon$ yields (3.5). □

Lemma 3.3. *Define*

$$E(\varepsilon) := (1-2^{-\varepsilon})\zeta(1+\varepsilon)=\sum_{n=0}^{\infty}a_n\varepsilon^n. \tag{3.6}$$

Then for every $n\geq 0$,

$$a_n=\sum_{r=0}^{n}\frac{(-1)^r\ell^{r+1}}{(r+1)!}\,\frac{\gamma_{n-r}}{(n-r)!}. \tag{3.7}$$

Proof. The factor $1-2^{-\varepsilon}$ has the Taylor expansion

$$1-2^{-\varepsilon}=1-e^{-\ell\varepsilon}=\sum_{m=1}^{\infty}\frac{(-1)^{m+1}\ell^m}{m!}\varepsilon^m=\sum_{r=0}^{\infty}\frac{(-1)^r\ell^{r+1}}{(r+1)!}\varepsilon^{r+1}.$$

The zeta factor has the Laurent expansion (3.1). The product (3.6) can be written as

$$E(\varepsilon)=\left(\sum_{r=0}^{\infty}\frac{(-1)^r\ell^{r+1}}{(r+1)!}\varepsilon^{r+1}\right)\left(\frac{1}{\varepsilon}+\sum_{j=0}^{\infty}\frac{(-1)^j\gamma_j}{j!}\varepsilon^j\right).$$

The coefficient of ε^n in $E(\varepsilon)$ arises from the products with $(r+1)+j=n+1$, that is $j=n-r$, and the $1/\varepsilon$ term contributes exactly one power of ε from the other factor. Thus

$$a_n=\sum_{r=0}^{n}\frac{(-1)^r\ell^{r+1}}{(r+1)!}\cdot\frac{(-1)^{n-r}\gamma_{n-r}}{(n-r)!}(-1)^{n-r}.$$

This simplifies to (3.7). □

Theorem 3.4 (Closed-form moments for arbitrary index)**.** *For every integer $k\geq 0$,*

$$\mu_k = k!\sum_{j=0}^{k}g_ja_{k-j}, \tag{3.8}$$

where g_j and a_{k-j} are given by (3.5) *and* (3.7)*. Equivalently,*

$$\boxed{\mu_k = k!\sum_{j=0}^{k}\frac{1}{j!}\mathrm{B}_j\big(1!c_1,2!c_2,\dots,j!c_j\big)\sum_{r=0}^{k-j}\frac{(-1)^r\ell^{r+1}}{(r+1)!}\,\frac{\gamma_{k-j-r}}{(k-j-r)!}} \tag{3.9}$$

with $c_1=-\gamma$, $c_m=(-1)^m\zeta(m)/m$ for $m\geq 2$, and $\ell=\log 2$.

Proof. Theorem 2.3 and Lemma 2.2 imply that

$$F(1+\varepsilon)=\sum_{k=0}^{\infty}\frac{\mu_k}{k!}\varepsilon^k.$$

On the other hand,

$$F(1+\varepsilon)=\Gamma(1+\varepsilon)\,(1-2^{-\varepsilon})\zeta(1+\varepsilon)=\left(\sum_{j=0}^{\infty}g_j\varepsilon^j\right)\left(\sum_{m=0}^{\infty}a_m\varepsilon^m\right).$$

By Cauchy's product formula, the coefficient of ε^k on the right is $\sum_{j=0}^{k} g_j a_{k-j}$. The coefficient of ε^k on the left is $\mu_k/k!$. This yields (3.8). Substituting (3.5) and (3.7) gives (3.9). □

4. Hankel determinants and Salem–Rao orthogonal polynomials

Definition 4.1. For $n \geq 0$, define the $(n+1) \times (n+1)$ Hankel matrix

$$H_n := \left(\mu_{i+j}\right)_{0 \leq i,j \leq n} = \begin{pmatrix} \mu_0 & \mu_1 & \cdots & \mu_n \\ \mu_1 & \mu_2 & \cdots & \mu_{n+1} \\ \vdots & \vdots & \ddots & \vdots \\ \mu_n & \mu_{n+1} & \cdots & \mu_{2n} \end{pmatrix}, \tag{4.1}$$

and the Hankel determinant

$$\Delta_n := \det H_n. \tag{4.2}$$

Set $\Delta_{-1} := 1$.

Theorem 4.2. *For every $n \geq 0$, the matrix H_n is positive definite, and*

$$\Delta_n > 0. \tag{4.3}$$

Proof. Let $c_0, \ldots, c_n \in \mathbb{R}$, not all zero, and define

$$q(u) := \sum_{j=0}^{n} c_j u^j.$$

Then

$$\sum_{i,j=0}^{n} c_i c_j \mu_{i+j} = \int_{\mathbb{R}} \left(\sum_{i=0}^{n} c_i u^i\right) \left(\sum_{j=0}^{n} c_j u^j\right) w(u)\ \mathrm{d}u = \int_{\mathbb{R}} q(u)^2 w(u)\ \mathrm{d}u.$$

The integrand $q(u)^2 w(u)$ is nonnegative on $\mathbb{R}$. If the integral equals zero, then $q(u)^2 w(u) = 0$ almost everywhere, and since $w(u) > 0$ for all $u \in \mathbb{R}$, it follows that $q(u) = 0$ for all $u \in \mathbb{R}$. A polynomial that vanishes identically on $\mathbb{R}$ has all coefficients zero, contradicting the assumption. Thus

$$\int_{\mathbb{R}} q(u)^2 w(u)\ \mathrm{d}u > 0$$

for every nonzero vector $(c_0, \ldots, c_n)$. This is exactly positive definiteness of H_n. A positive definite matrix has strictly positive determinant, so $\Delta_n > 0$. □

Theorem 4.3 (Determinantal formula for the Salem–Rao polynomials)**.** *For each $n \geq 0$ there exists a unique monic polynomial P_n of degree n such that*

$$\langle P_n, u^m \rangle_w = 0, \qquad 0 \leq m \leq n-1. \tag{4.4}$$

It is given by

$$P_n(u) = \frac{1}{\Delta_{n-1}} \begin{vmatrix} \mu_0 & \mu_1 & \cdots & \mu_n \\ \mu_1 & \mu_2 & \cdots & \mu_{n+1} \\ \vdots & \vdots & \ddots & \vdots \\ \mu_{n-1} & \mu_n & \cdots & \mu_{2n-1} \\ 1 & u & \cdots & u^n \end{vmatrix}, \tag{4.5}$$

with $\Delta_{-1} := 1$.

Proof. Fix $n \geq 0$. Consider the ansatz

$$P_n(u) = u^n + \sum_{j=0}^{n-1} p_{n,j} u^j.$$

The orthogonality conditions (4.4) yield

$$\langle P_n, u^m \rangle_w = \mu_{n+m} + \sum_{j=0}^{n-1} p_{n,j} \mu_{j+m} = 0, \qquad 0 \leq m \leq n-1.$$

This is a linear system in the unknowns $p_{n,0}, \ldots, p_{n,n-1}$ with coefficient matrix H_{n-1} and right-hand side $-(\mu_n, \mu_{n+1}, \ldots, \mu_{2n-1})^\top$. The matrix H_{n-1} is positive definite by Theorem 4.2, hence invertible, so the system has a unique solution and P_n exists and is unique.

Cramer's rule yields the coefficients $p_{n,j}$ as ratios of determinants, obtained by replacing the j-th column of H_{n-1} by the right-hand side. Assembling these into a single formula with the last row $(1, u, \ldots, u^n)$ gives (4.5). Expansion along the last row shows that the coefficient of u^n is

$$\frac{\Delta_{n-1}}{\Delta_{n-1}} = 1,$$

so P_n is monic. □

Theorem 4.4 (Squared norms for the Salem–Rao polynomials)**.** *Let*

$$h_n := \langle P_n, P_n \rangle_w .$$

Then for every $n \geq 0$,

$$h_n = \frac{\Delta_n}{\Delta_{n-1}}. \tag{4.6}$$

Proof. For each $n \geq 0$, orthogonality implies that P_n is orthogonal to every polynomial of degree at most $n-1$. Since $u^n - P_n$ has degree at most $n-1$, one has

$$\langle u^n - P_n, P_n \rangle_w = 0$$

and therefore

$$h_n = \langle P_n, P_n \rangle_w = \langle u^n, P_n \rangle_w .$$

Substitute (4.5) into this expression. The last row $(1, u, \ldots, u^n)$ in (4.5), after taking the inner product with u^n, becomes $(\mu_n, \mu_{n+1}, \ldots, \mu_{2n})$, because $\langle u^m, u^n \rangle_w = \mu_{m+n}$. Thus

$$h_n = \frac{1}{\Delta_{n-1}} \begin{vmatrix} \mu_0 & \mu_1 & \cdots & \mu_n \\ \mu_1 & \mu_2 & \cdots & \mu_{n+1} \\ \vdots & \vdots & \ddots & \vdots \\ \mu_{n-1} & \mu_n & \cdots & \mu_{2n-1} \\ \mu_n & \mu_{n+1} & \cdots & \mu_{2n} \end{vmatrix} = \frac{\Delta_n}{\Delta_{n-1}}.$$

□

5. Three-term recurrence and closed recurrence coefficients

Theorem 5.1 (Salem–Rao Favard recurrence)**.** *The monic Salem–Rao polynomials $\{P_n\}_{n \geq 0}$ satisfy*

$$P_{n+1}(u) = (u - \alpha_n) P_n(u) - \beta_n P_{n-1}(u), \qquad n \geq 0, \tag{5.1}$$

with $P_{-1} \equiv 0$, $P_0 \equiv 1$, and recurrence coefficients

$$\alpha_n = \frac{\langle u P_n, P_n \rangle_w}{h_n}, \qquad n \geq 0, \tag{5.2}$$

$$\beta_n = \frac{h_n}{h_{n-1}}, \qquad n \geq 1. \tag{5.3}$$

In particular, $\beta_n > 0$ *for all* $n \geq 1$.

Proof. Fix $n \geq 0$. The polynomial uP_n has degree $n+1$ and leading coefficient 1. Hence $uP_n - P_{n+1}$ has degree at most n and therefore belongs to $\operatorname{span}\{P_0, \ldots, P_n\}$. There exist unique real numbers $\alpha_n, \beta_n, \gamma_{n,0}, \ldots, \gamma_{n,n-2}$ such that

$$uP_n(u) = P_{n+1}(u) + \alpha_n P_n(u) + \beta_n P_{n-1}(u) + \sum_{j=0}^{n-2} \gamma_{n,j} P_j(u), \tag{5.4}$$

where, for $n = 0$, the terms P_{n-1} and the sum are absent and α_0 is the only coefficient.

Taking the inner product of (5.4) with P_n yields

$$\langle uP_n, P_n \rangle_w = \alpha_n \langle P_n, P_n \rangle_w = \alpha_n h_n,$$

because P_{n+1} and all P_j with $j \leq n-1$ are orthogonal to P_n. Thus (5.2) holds.

For $n \geq 1$, taking the inner product of (5.4) with P_{n-1} yields

$$\langle uP_n, P_{n-1} \rangle_w = \beta_n \langle P_{n-1}, P_{n-1} \rangle_w = \beta_n h_{n-1},$$

because P_{n+1} and P_n are orthogonal to P_{n-1} and $\langle P_j, P_{n-1} \rangle_w = 0$ for $j \leq n-2$ as well. Symmetry of the inner product gives $\langle uP_n, P_{n-1} \rangle_w = \langle P_n, uP_{n-1} \rangle_w$.

The polynomial uP_{n-1} has degree n and leading coefficient 1, and therefore has the form

$$uP_{n-1}(u) = P_n(u) + \widetilde{\alpha}_{n-1} P_{n-1}(u) + \sum_{j=0}^{n-2} \widetilde{\gamma}_{n-1,j} P_j(u) \tag{5.5}$$

for some real numbers $\widetilde{\alpha}_{n-1}, \widetilde{\gamma}_{n-1,j}$. Taking the inner product with P_n gives

$$\langle uP_{n-1}, P_n \rangle_w = \langle P_n, P_n \rangle_w = h_n,$$

because P_n is orthogonal to P_{n-1} and all lower-degree polynomials P_j. Thus

$$\langle uP_n, P_{n-1} \rangle_w = \langle uP_{n-1}, P_n \rangle_w = h_n,$$

and therefore $\beta_n h_{n-1} = h_n$, which proves (5.3).

Substituting (5.2) and (5.3) into (5.4) and rearranging yields (5.1). The positivity of h_n (Theorem 4.4) and h_{n-1} implies $\beta_n > 0$ for all $n \geq 1$. □

Corollary 5.2 (Closed determinant formula for β_n)**.** *For every* $n \geq 1$,

$$\beta_n = \frac{\Delta_n \, \Delta_{n-2}}{\Delta_{n-1}^2}. \tag{5.6}$$

Proof. Combination of (5.3) with (4.6) gives

$$\beta_n = \frac{\Delta_n / \Delta_{n-1}}{\Delta_{n-1} / \Delta_{n-2}} = \frac{\Delta_n \, \Delta_{n-2}}{\Delta_{n-1}^2}.$$

□

Theorem 5.3 (Closed determinant formula for α_n)**.** *For every* $n \geq 0$,

$$\alpha_n = \frac{\det A_n}{\Delta_n}, \tag{5.7}$$

where the $(n+1)\times(n+1)$ *matrix* A_n *is*

$$A_n := \begin{pmatrix} \mu_0 & \mu_1 & \cdots & \mu_{n-1} & \mu_{n+1} \\ \mu_1 & \mu_2 & \cdots & \mu_n & \mu_{n+2} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \mu_n & \mu_{n+1} & \cdots & \mu_{2n-1} & \mu_{2n+1} \end{pmatrix}. \tag{5.8}$$

Proof. By (5.2) and (4.6),

$$\alpha_n = \frac{\langle uP_n, P_n\rangle_w}{h_n} = \frac{\Delta_{n-1}}{\Delta_n}\langle uP_n, P_n\rangle_w .$$

Substitute the determinantal representation $P_n(u) = \Delta_{n-1}^{-1} D_n(u)$, where

$$D_n(u) := \begin{vmatrix} \mu_0 & \mu_1 & \cdots & \mu_n \\ \mu_1 & \mu_2 & \cdots & \mu_{n+1} \\ \vdots & \vdots & \ddots & \vdots \\ \mu_{n-1} & \mu_n & \cdots & \mu_{2n-1} \\ 1 & u & \cdots & u^n \end{vmatrix}.$$

Then

$$\langle uP_n, P_n\rangle_w = \frac{1}{\Delta_{n-1}^2}\langle uD_n, D_n\rangle_w .$$

Thus

$$\alpha_n = \frac{1}{\Delta_n}\langle uD_n, D_n\rangle_w .$$

It suffices to identify $\langle uD_n, D_n\rangle_w$ with $\det A_n$. Expansion of $D_n(u)$ along the last row gives

$$D_n(u) = \sum_{j=0}^{n} (-1)^{n+j} C_{n,j} u^j,$$

where $C_{n,j}$ is the $(n+1, j+1)$-cofactor of the matrix in the definition of $D_n(u)$. Therefore

$$uD_n(u) = \sum_{j=0}^{n} (-1)^{n+j} C_{n,j} u^{j+1}.$$

Write

$$uD_n(u) = \sum_{m=0}^{n+1} \widetilde{C}_{n,m} u^m$$

with $\widetilde{C}_{n,0} = 0$ and $\widetilde{C}_{n,m} = (-1)^{n+m-1} C_{n,m-1}$ for $1 \le m \le n+1$. Then

$$\langle uD_n, D_n\rangle_w = \sum_{m=0}^{n+1}\sum_{j=0}^{n} \widetilde{C}_{n,m}(-1)^{n+j} C_{n,j} \left\langle u^m, u^j\right\rangle_w = \sum_{m=0}^{n+1}\sum_{j=0}^{n} \widetilde{C}_{n,m}(-1)^{n+j} C_{n,j}\mu_{m+j}.$$

By multilinearity of determinants and the definition of cofactors, this double sum is exactly the determinant of the matrix A_n obtained from the Hankel matrix H_n by replacing its last column $(\mu_n, \ldots, \mu_{2n})^\top$ with $(\mu_{n+1}, \ldots, \mu_{2n+1})^\top$. Explicitly,

$$\langle uD_n, D_n\rangle_w = \det A_n.$$

Thus $\alpha_n = \det A_n/\Delta_n$, which is (5.7). □

6. Christoffel–Darboux identity and zeros

Theorem 6.1 (Christoffel–Darboux identity)**.** *For every $n \geq 0$ and all $u, v \in \mathbb{R}$ with $u \neq v$,*

$$\sum_{j=0}^{n} \frac{P_j(u)P_j(v)}{h_j} = \frac{P_{n+1}(u)P_n(v) - P_n(u)P_{n+1}(v)}{h_n(u-v)}. \tag{6.1}$$

Proof. For each $j \geq 0$, the recurrence (5.1) gives

$$uP_j(u) = P_{j+1}(u) + \alpha_j P_j(u) + \beta_j P_{j-1}(u),$$
$$vP_j(v) = P_{j+1}(v) + \alpha_j P_j(v) + \beta_j P_{j-1}(v).$$

Multiply the first equality by $P_j(v)$ and the second by $P_j(u)$, then subtract:

$$\begin{aligned}(u-v)P_j(u)P_j(v) &= P_{j+1}(u)P_j(v) - P_j(u)P_{j+1}(v) \\ &\quad + \beta_j\big(P_{j-1}(u)P_j(v) - P_j(u)P_{j-1}(v)\big).\end{aligned}$$

Divide by h_j. The relation $\beta_j = h_j/h_{j-1}$ (for $j \geq 1$) implies

$$\frac{\beta_j}{h_j} = \frac{1}{h_{j-1}}.$$

Thus

$$\begin{aligned}(u-v)\frac{P_j(u)P_j(v)}{h_j} &= \frac{P_{j+1}(u)P_j(v) - P_j(u)P_{j+1}(v)}{h_j} \\ &\quad - \frac{P_j(u)P_{j-1}(v) - P_{j-1}(u)P_j(v)}{h_{j-1}}.\end{aligned}$$

Summation from $j = 0$ to n produces a telescoping sum on the right-hand side. The $j = 0$ lower term vanishes because $P_{-1} \equiv 0$ and h_{-1} is not used. The remaining terms collapse to

$$(u-v)\sum_{j=0}^{n} \frac{P_j(u)P_j(v)}{h_j} = \frac{P_{n+1}(u)P_n(v) - P_n(u)P_{n+1}(v)}{h_n}.$$

Division by $u - v$ yields (6.1). □

Theorem 6.2 (Zeros of Salem–Rao polynomials)**.** *For every $n \geq 1$, the Salem–Rao polynomial P_n has exactly n distinct real zeros. Moreover, the zeros of P_n and P_{n+1} interlace: between any two consecutive zeros of P_{n+1} there lies exactly one zero of P_n, and between any two consecutive zeros of P_n there lies exactly one zero of P_{n+1}.*

Proof. Consider P_n. Orthogonality with respect to a strictly positive measure on $\mathbb{R}$ implies that P_n has at most n distinct real zeros. To prove that it has at least n sign changes, suppose that the number of distinct real zeros of P_n in $\mathbb{R}$ at which the sign changes is $m < n$. Let $u_1 < \cdots < u_m$ denote these zeros and define

$$q(u) := \prod_{k=1}^{m}(u - u_k).$$

The polynomial q has degree $m < n$ and the product $P_n(u)q(u)$ is nonnegative or nonpositive on $\mathbb{R}$ and not identically zero. The inner product $\langle P_n, q\rangle_w$ equals

$$\int_{\mathbb{R}} P_n(u)q(u)w(u)\ \mathrm{d}u,$$

which is strictly positive or strictly negative, because $w(u) > 0$ for all $u \in \mathbb{R}$ and $P_n(u)q(u)$ does not change sign and is not almost everywhere zero. On the other hand, $\deg q < m < n$, so orthogonality implies $\langle P_n, q\rangle_w = 0$, a contradiction. Therefore P_n has at least n sign changes, hence at least n

distinct real zeros. A polynomial of degree n cannot have more than n distinct zeros, so P_n has exactly n distinct real zeros, all simple.

Interlacing follows from the Christoffel–Darboux identity. Let $x_1 < \cdots < x_n$ denote the distinct real zeros of P_n. For any real number u, the identity (6.1) with $v = u$ and a limiting argument yields

$$\sum_{j=0}^{n} \frac{P_j(u)^2}{h_j} = \frac{P'_{n+1}(u)P_n(u) - P'_n(u)P_{n+1}(u)}{h_n}. \tag{6.2}$$

The left-hand side is a sum of nonnegative terms and is strictly positive, since $P_0 \equiv 1$ and $h_0 > 0$. Consequently, the numerator on the right-hand side never vanishes. At a zero $u = x_k$ of P_n, one has $P_n(x_k) = 0$ and $P'_n(x_k) \neq 0$ (zeros are simple), so

$$\sum_{j=0}^{n} \frac{P_j(x_k)^2}{h_j} = -\frac{P'_n(x_k)P_{n+1}(x_k)}{h_n}.$$

The left-hand side is strictly positive, hence the product $-P'_n(x_k)P_{n+1}(x_k)$ is positive. Thus $P'_n(x_k)$ and $P_{n+1}(x_k)$ have opposite signs. Since P_n alternates sign at successive zeros x_k, the values $P_{n+1}(x_k)$ alternate sign as well.

The intermediate value theorem then implies that P_{n+1} has at least one zero between x_k and x_{k+1} for each $k = 1, \ldots, n-1$. This furnishes at least $n-1$ zeros of P_{n+1} in the open intervals between successive zeros of P_n. Additional consideration of the signs of P_n and P_{n+1} at $\pm\infty$ shows that there is one zero of P_{n+1} to the left of x_1 and one to the right of x_n, yielding $n+1$ distinct real zeros of P_{n+1}. Since a polynomial of degree $n+1$ has at most $n+1$ distinct zeros, these are all the real zeros of P_{n+1}, and each interval between successive zeros of one polynomial contains exactly one zero of the other. This is interlacing. □

7. Summary of closed formulas for the Salem–Rao system

The construction for the Salem–Rao weight $w(u) = e^u/(e^{e^u} + 1)$ yields the following closed formulas:

- For each $k \geq 0$, the moment μ_k is given by the finite double sum (3.9) in terms of the Euler–Mascheroni constant, the Stieltjes constants, the values $\zeta(m)$, and complete exponential Bell polynomials.
- For each $n \geq 0$, the monic Salem–Rao polynomial P_n is given by the Hankel determinant (4.5), whose entries are the moments μ_k.
- For each $n \geq 0$, the squared $L^2(w(u)\,\mathrm{d}u)$ norm h_n is given by (4.6) as the ratio of Hankel determinants Δ_n/Δ_{n-1}.
- For each $n \geq 1$, the recurrence coefficient β_n is given by (5.6) in terms of Δ_n, Δ_{n-1}, and Δ_{n-2}:
$$\beta_n = \frac{\Delta_n\,\Delta_{n-2}}{\Delta_{n-1}^2}.$$
- For each $n \geq 0$, the recurrence coefficient α_n is given by (5.7) as the ratio $\det A_n/\Delta_n$, where A_n is defined in (5.8).

In particular, for the Salem–Rao system, the three-term recurrence (5.1) is fully solved in the sense that both coefficients α_n and β_n are explicit closed functions of the index n and the data $\{\zeta(m)\}_{m\geq 2}$, $\{\gamma_m\}_{m\geq 0}$ and ℓ.

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